

Coupling Interpretable Allocation, Model-Predictive Control, and Tail-Aware Forecasting for Live Portfolio Management

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Outline

Live portfolio management couples three problems: allocation, the timing of rebalancing, and evaluation beyond the historical record.

1 Allocation — What to hold. Maximize utility of a score γ for each stock, within trading limits.

2 Rebalancing — When to trade. Re-plan at the 21-day horizon, a band exit, or a drawdown stop; real-data backtest, 2025–26.

3 Market generator — Evaluation. A backtest is one historical path; JumpHMM generates typical and stressed futures.

We want to build evolving portfolios that maximize investor satisfaction as preferences shift with the market.

Maximize utility subject to trading limits (convex cone)

Each rebalance chooses n_i shares of each name i to solve:

$$\max_{n \geq 0} \; U(n;\gamma)$$ utility U (γ_i: score of name i, next slide)

s.t. $\sum_i n_i S_i \leq W$ budget W (S_i: price of name i)

$w_i \leq \bar{w}$ single-name cap $\bar{w}$ (w_i: weight of name i)

$\sqrt{w^\top \Sigma_g \, w} \leq \sigma_{\text{SPY}}$ volatility cap σ_{SPY} (Σ_g: growth-rate covariance)

$\bar{c} \, \|n - n^{\text{prev}}\|_1 \leq B$ turnover budget B ($\bar{c}$: half spread per share)

$\beta^\top w \leq \bar{\beta}$ β cap $\bar{\beta}$, hedge on (β: beta to SPY)

	Cobb–Douglas	CES on shares	CES on weights	Stone–Geary
$U(n;\gamma)$, over $\gamma_i > 0$	$\sum \gamma_i \ln n_i$	$(\sum \gamma_i n_i^\rho)^{1/\rho}$	$(\sum \gamma_i w_i^\rho)^{1/\rho}$	$\sum \gamma_i \ln(n_i - b_i)$
budget alone	$w_i \propto \gamma_i$	$n_i \propto (\gamma_i / S_i)^\eta$	$w_i \propto \gamma_i^\eta$	$(n_i - b_i)\, S_i \propto \gamma_i$

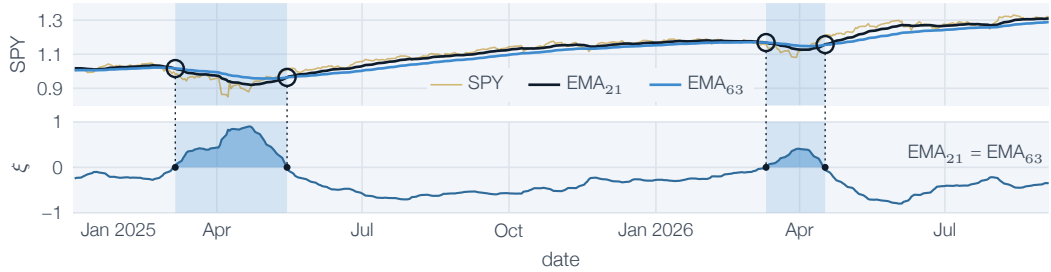
$\rho = 1 - 1/\eta$, $\eta = 0.5$; $b_i S_i = 1\%$ of W. Also compared: minimum variance, maximum Sharpe, 130/30 long-short.

The score γ: expected growth, tilted by the market

$$\gamma_i = \tanh\left(\frac{\alpha_i + \beta_i\,\tilde{g}_M}{\max(|\beta_i|,\,0.1)^{\xi}}\right)$$

expected growth (α_i, β_i: fit of name i to SPY; $\tilde{g}_M$: SPY's growth rate, 252-day EMA)

β penalty (ξ: market state, >0 when $\text{EMA}_{21} < \text{EMA}_{63}$)



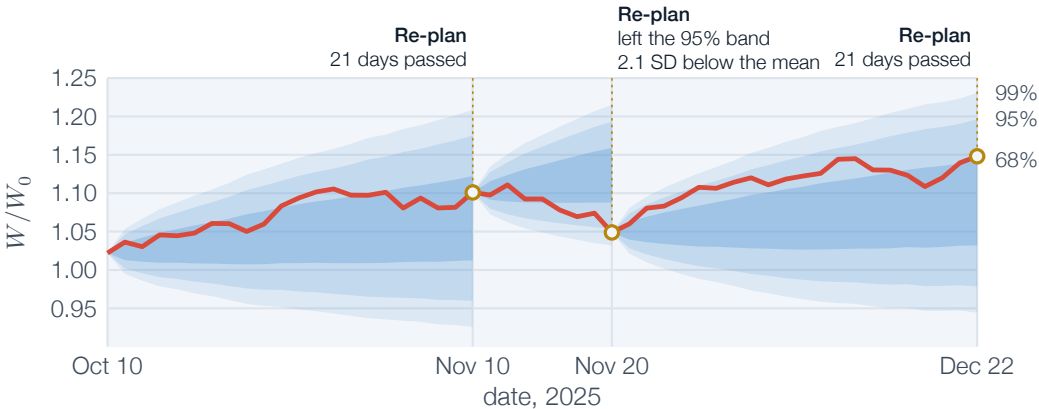
Weighting	γ enters $U(n;\gamma)$: names with $\gamma_i > 0$ are held long, and 130/30 shorts $\gamma_i < 0$.
Selection	the 13 highest γ overall, or the 3 highest in each sector.

$\xi = -G\,(\text{EMA}_{21}/\text{EMA}_{63} - 1)$ from SPY's 21- and 63-day exponential averages; $|\xi| \le 1$ on 99% of 2014–2024 days.

Markets drift away from any plan, so when should the portfolio stop and re-plan?

Rebalancing triggers: horizon, forecast band, drawdown

Forecast: multi-asset GBM, 1,000 paths, 21 trading days ahead. Plant: real prices.



Hand-picked 13, Cobb–Douglas, hedge off; over Jan 2025 – Sep 2026, 27 re-plans, 10 before the horizon.

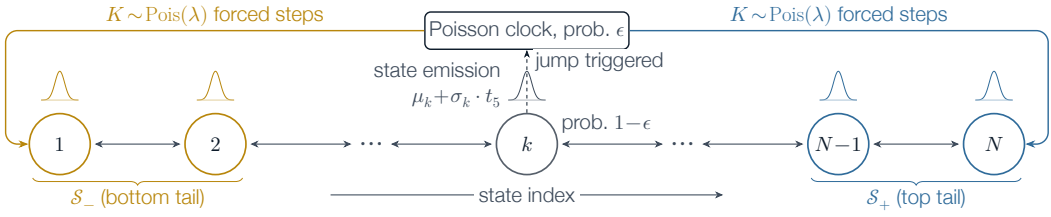
Stock selection matters more than the weighting rule

Real-data backtest, Jan 2025 – Sep 2026, 373 names. Each row is a selection rule.



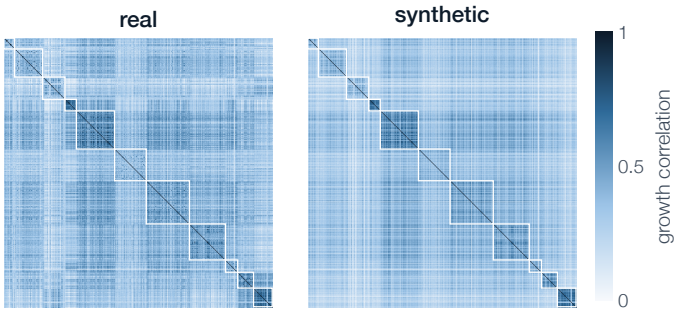
What if 2025–26 was just a good run? Let's test our approaches on fat-tailed, multi-asset synthetic markets.

JumpHMM: a market generator with fat tails



Synthetic names move together like the real ones

$$g_i(t) = \underbrace{\alpha_i}_{\text{trend}} + \underbrace{\beta_i(t)\, g_m(t)}_{\text{market: 1 JumpHMM}} + \underbrace{\delta_i\, f_s(t)}_{\text{sector: 11 JumpHMMs}} + \underbrace{s_i\, \tilde{\varepsilon}_i(t)}_{\text{own: 373 JumpHMMs}}$$



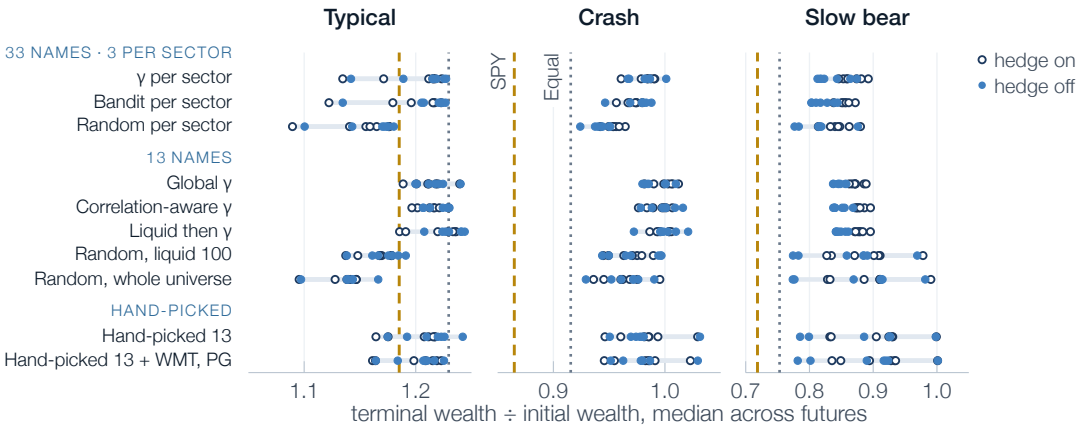
mean pairwise correlation

	within a sector	across sectors
real	0.47	0.35
synthetic	0.45	0.31

Real: 373 names, 2014–2024. Synthetic: one simulated path, same dates. Names sorted by sector; sectors outlined.

Simulated sell-offs: every median ends above SPY

Of 10,000 futures: 600 typical (random), plus the 200 worst SPY drops of each kind.



Crash: months of extreme up and down days. Slow bear: no such stretch, a steady decline.

Key takeaways

- **Allocation: stock selection mattered more than the weighting rule.**
  With the hedge off, seven weighting rules on one 33-name basket ended at 1.09–1.23 times initial wealth; different baskets ended at 1.03–1.70.

- **Rebalancing: re-plans came at the horizon, a band exit, or a drawdown stop.**
  About 27 re-plans in 419 trading days, a third of them before the 21-day horizon.

- **Evaluation: in both sell-off sets, every median ended above SPY.**
  JumpHMM reproduces fat tails, but its volatility clustering fades after a day. With the hedge off, 13 names picked by γ beat 13 random names in 62–71% of typical futures; in slow-bear futures, random picks did as well or better for most rules.

A paper-trading version of this system is running now; its results are not part of this talk.

Acknowledgements

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Evan Vu, class of 2026, now at Bridgewater
an early version of the Cobb–Douglas allocation and the bandit picker

We also thank Alpaca Markets, and Polygon.io for the original daily price data.

Paper

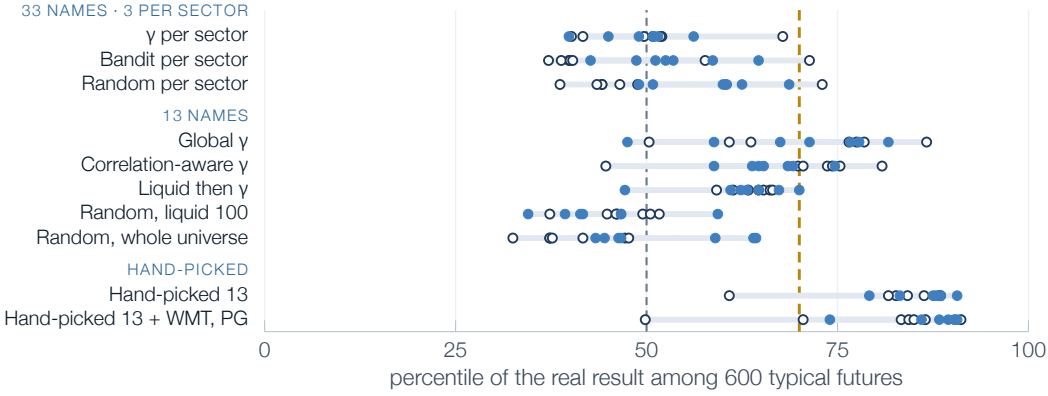
arXiv:2611.03022

Code

github.com/varnerlab/INFORMS-Annual-Meeting-2026-Varnerlab

Real results as percentiles of typical futures

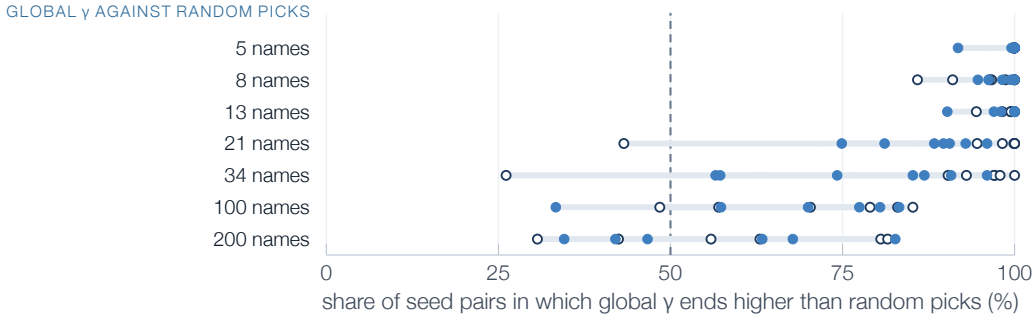
Each real result among 600 typical JumpHMM futures. Right of 50 beats the median future.



Real results use the simulated runs' seeds: 5660–5664 for random and bandit pickers, 5660 for the rest.

Global γ against random picks, 5 to 200 names

The K highest γ in the universe against K random names, from 5 to 200 names. At 50%, the two picks end higher equally often.



● hedge off ○ hedge on one dot per weighting rule

Real-data backtest, 30 runs each. Stone–Geary left out at 100 and 200 names: its 1%-per-name floor uses the whole budget at 100 names and twice the budget at 200.

Terminal wealth by picker and utility: hedge off

Picker	CD	CES	CESw	SG	MinVar	Tangent	LS
γ per sector	1.230	1.219	1.221	1.234	1.086	1.227	1.110
Bandit per sector	1.226	1.234	1.223	1.239	1.162	1.229	1.140
Random per sector	1.205	1.183	1.201	1.183	1.128	1.158	1.127
Global γ	1.520	1.384	1.460	1.441	1.355	1.341	1.248
Correlation-aware γ	1.352	1.386	1.421	1.380	1.284	1.404	1.219
Liquid then γ	1.333	1.310	1.379	1.350	1.238	1.361	1.305
Random, liquid 100	1.216	1.173	1.211	1.212	1.158	1.274	1.152
Random, whole universe	1.098	1.132	1.138	1.112	1.104	1.160	1.031
Hand-picked 13	1.548	1.571	1.614	1.631	1.443	1.697	1.549
Hand-picked 13 + WMT, PG	1.616	1.646	1.637	1.639	1.362	1.642	1.622
SPY buy-and-hold	1.317	1.317	1.317	1.317	1.317	1.317	1.317
equal-weight universe	1.296	1.296	1.296	1.296	1.296	1.296	1.296

Terminal wealth ÷ initial wealth, median of 30 runs. Real-data backtest, Jan 2025 – Sep 2026.
CD Cobb–Douglas · CES on shares · CESw CES on weights · SG Stone–Geary · MinVar minimum variance · Tangent the tangency portfolio · LS 130/30 long-short Cobb–Douglas.

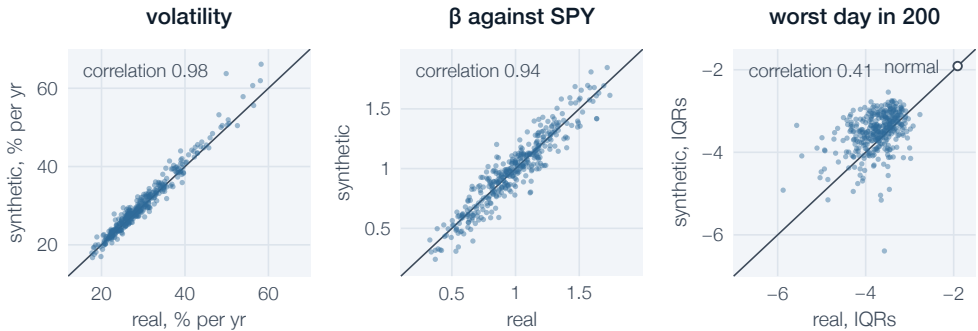
Terminal wealth by picker and utility: hedge on

Picker	CD	CES	CESw	SG	MinVar	Tangent	LS
γ per sector	1.212	1.232	1.223	1.218	1.073	1.285	1.130
Bandit per sector	1.125	1.140	1.167	1.194	1.166	1.214	1.120
Random per sector	1.108	1.087	1.114	1.155	1.155	1.092	1.092
Global γ	1.503	1.305	1.464	1.461	1.203	1.589	1.283
Correlation-aware γ	1.409	1.429	1.479	1.447	1.213	1.366	1.292
Liquid then γ	1.322	1.285	1.347	1.331	1.184	1.317	1.336
Random, liquid 100	1.167	1.156	1.174	1.178	1.123	1.155	1.112
Random, whole universe	1.069	1.106	1.093	1.115	1.079	1.062	0.994
Hand-picked 13	1.554	1.517	1.517	1.511	1.233	1.582	1.586
Hand-picked 13 + WMT, PG	1.564	1.571	1.543	1.542	1.159	1.383	1.630
SPY buy-and-hold	1.317	1.317	1.317	1.317	1.317	1.317	1.317
equal-weight universe	1.296	1.296	1.296	1.296	1.296	1.296	1.296

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Each synthetic name resembles its real counterpart

$$g_i(t) = \underbrace{\alpha_i}_{\text{trend}} + \underbrace{\beta_i(t)\, g_m(t)}_{\text{market: 1 JumpHMM}} + \underbrace{\delta_i\, f_s(t)}_{\text{sector: 11 JumpHMMs}} + \underbrace{s_i\, \tilde{\varepsilon}_i(t)}_{\text{own: 373 JumpHMMs}}$$



Each dot is a name, real 2014–2024 against one simulated path. IQR: interquartile range of daily growth.